Micro I, class 26

After considering a differentiated-Bertrand game as an example of a SSPm game, I introduced repeated games, and I asserted two facts about a finitely repeated prisoners' dilemma: the unique SPNE is that each player's strategy is "defect at each date t, *no matter what the history*;" and that the unique SPNE *outcome* is "(D,D) at each date." The first we proved by induction on the length τ from the end of the game; the second I will prove by contraposition next week: if a strategy profile $\tilde{s}$ generates a different outcome, then it is not a Nash equilibrium.

It turns out that both of these results can be generalized: if a stage game has a unique NE, then any finite repetition of it has a unique SPNE;[1] if, in addition, each player's equilibrium utility in the stage game equals its *minimax* utility, then any finite repetition of it has a unique NE outcome.[2] (A player's minimax utility in a game $G = (N, \{A_i\}_{i\in N}, \{\pi_i\}_{i\in N})$ is $v_i = \min_{a_{-i}\in A_{-i}}[\max_{a_i\in A_i} \pi_i(a)]$.)

The second result need not generalize to other games. Consider the two-player Cournot game played twice, with $p(Q) = \max\{0, A - Q\}$ and $c_i(q_i) = kq_i$, with $A > k > 0$. With no discounting (or δ close enough to 1), this profile is a Nash equilibrium: firm 1 produces its monopoly at date 1, $(A - k)/2$, and if firm 2 produces positive output at date 1, firm 1 produces output of A at date 2; otherwise firm 1 produces the one-shot Cournot output at date 2, $(A - k)/3$. A best response for firm 2 is to choose $q_2 = 0$ at date 1, and to produce its one-shot Cournot output at date 2, no matter what the history. This gives firm 2 zero profit at date 1 and its one-shot Cournot profit at date 2. If it produces a positive output at date 1, it earns *less than* its one-shot Cournot profit at date 1 and zero profit at date 2, no matter what its date-2 strategy component is. Provided that it does not discount too much, the second discounted profit stream is less than the first. Of course the Nash equilibrium is not subgame perfect.

I then turned to the infinitely repeated prisoners' dilemma. We confirmed that the grim trigger strategy profile (in which each player chooses C after a history *if and only if* the history consists of all C's), provided that the players are patient enough.

Next time we will extend the last result to a broader class of games, and then I will discuss several examples, including the repeated Bertrand game. In what follows I outline the argument from Tuesday about SPNE of a finitely- and an infinitely-repeated prisoners' dilemma.

[1] I give the argument in what follows

[2] A proof can be adapted from our argument for the prisoners' dilemma.

$SPNE(G^T_{PD})$ for T finite, $\delta = 1$.

Fix $T \geq 2$. I will use induction on the number of dates τ until the end of the game. Let $\tau = 0$, and consider the subgame starting at any $h_{T-\tau-1} = h_{T-1} \in H_{T-1}$. The unique NE of this subgame is (D, D). Now let $\tau \geq 1$, consider the subgame starting at any $h_{T-\tau-1} \in H_{T-\tau-1}$, and suppose that $f^t_i(h_{t-1}) = D$ for $i = 1, 2$ and $t \geq T - \tau + 1$. The outcome from $t = T - \tau + 1$ to T is therefore "(D, D) at every date $t \geq T - \tau + 1$, no matter what the players choose at $h_{T-\tau-1}$." The utility starting at $h_{T-\tau-1}$ is

$$\pi_i(a_1, a_2) + \tau \times \pi_i(D, D) = \pi_i(a_1, a_2) + 0.$$

The unique NE of this game—with strategy sets equal to $\{C, D\}$—is (D, D). It follows that the unique SPNE of a finitely-repeated prisoners' dilemma is that each player i chooses D *after every history*: for $T < \infty$, $SPNE(G^T_{PD}) = \{(s^*_1, s^*_2)\}$, where, for $i = 1, 2$, s^*_i is given by $f^{t*}_i(h_{t-1}) = D$ for every $h_{t-1} \in H_{t-1}$ and for $t = 1, ..., T$.

Two SPNE for $G^\infty_{PD}(\delta)$, with $0 < \delta < 1$.

First, consider a strategy $\hat{s}_i$ given by $\hat{f}^t_i(h_{t-1}) = D$ for every $h_{t-1} \in H_{t-1}$ and $t = 1, 2, 3, ...$ The profile $\hat{s}$ is certainly a NE for $G^\infty_{PD}(\delta)$ for any $0 < \delta < 1$: if one player choose "D after every history," then the other player can do no better than this strategy.

Now consider the grim trigger strategy s^*_i defined by, for $t = 1, 2, ...,$ $f^{t*}_i(h_{t-1}) = C$ if $h_{t-1} = h_0CC...CC$; and $f^{t*}_i(h_{t-1}) = D$ otherwise. Now suppose that $s_2 = s^*_2$. Then $u_1(s^*) = \frac{2}{1-\delta}$. It turns out that s^*_1 is a best response to s^*_2 iff $u_1(\hat{s}_1, s^*_2) \leq u_1(s^*_1, s^*_2)$: in order to do better, player 1's strategy must choose D along the path of play, then choose D forever after; and if it is not better to do it at date 1, it can't be better to do it at any other date. The grim trigger is a best response to the grim trigger iff (using the utility numbers from class)

$$\frac{2}{1-\delta} \geq 3 + 0 \times \frac{\delta}{(1-\delta)} \quad \text{or} \quad \delta \geq \frac{1}{3}.$$

This strategy profile is also subgame perfect if this inequality is met. There are two classes of subgames date-t subgames: A) both players has chosen C in in the past: $h_{t-1} = h_0CC....CC$; and B) all other date-$t-1$ histories. For A), the grim trigger strategies imply that the players should choose C at h_{t-1}, and C in the future if and only if no one has chosen D in the past. Provided that $\delta \geq 1/3$, the grim trigger strategy profile induces a Nash equilibrium on each such subgame, by the previous calculation. For B), the strategies indicate each player should choose D at h_{t-1} and after *any* subsequent history. We have already argued that this strategy profile is a Nash equilibrium of an infinitely repeated prisoners' dilemma.